Nathan DeBenedictis, Ben Hyde, Siham Ibrahim, and Amanda Zhong
SDS 4425: Data Mining Methods and Applications
Dr. Carlos Misael Madrid Padilla
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Project Proposal: Separating Malignant and Benign Breast Tumors Using Dimensionality Reduction, Clustering, and Regularized Classification

Research Question

We plan to investigate whether measurements of cell nuclei from breast biopsy images can reliably distinguish malignant tumors from benign ones, while simultaneously determining which measurements matter the most. More specifically, our research questions are: (1) Do the malignant and benign tumors naturally separate into two groups that match the diagnosis, even when the labels are hidden? (2) How do regularized classification models perform in predicting malignancy and which features do they rely on? As breast cancer is one of the most common cancers worldwide, and fine needle aspiration (FNA) is a quick and minimally invasive diagnostic test, a data-driven model like this could help support pathologists' decision-making processes. More specifically, since a false negative in missing a cancer diagnosis is important to identify than a false alarm, we will prioritize correctly identifying malignant tumors on top of assessing overall accuracy.

Dataset

To investigate our research questions, we will use the Breast Cancer Wisconsin Diagnostic dataset (Wolberg, Mangasarian, Street, & Street, 1993), which is available from the UCI Machine Learning Repository (https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic) and also built into scikit-learn as sklearn.datasets.load_breast_cancer (https://scikit-learn.org/stable/modules/generated/sklearn.datasets.load_breast_cancer.html). This dataset contains 569 observations, which are each a digitized image of an FNA sample from a breast mass. Each observation is labeled malignant (212) or benign (357) and has measurements for 10 properties of the cell nuclei (radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, and fractal dimension. From there, each of the 10 properties is summarized across all nuclei in the image by its mean, standard error, and "worst" value (the mean of the three largest), resulting in a total of 30 continuous features. While the dataset has no missing values, data preprocessing will include relabeling the data so that malignant = 1 and benign = 0. Additionally, we will standardize the features, so they are on the same scale. When splitting the data into training and test sets, we will keep the same ratio of malignant to benign tumors in each. Lastly, we will check which features are strongly correlated since some features such as radius, perimeter, and area essentially measure the same properties.

Planned methods

To begin, we will use summary statistics, histograms, boxplots, and a correlation heatmap to compare features between malignant and benign tumors. This will show which measurements differ the most and which are highly correlated. Since many features are correlated, PCA will reduce the 30 features to a few components that capture most of the variation. Plotting the first two components by diagnosis will show whether the two groups separate visually. Next, we will cluster the data into groups without using the labels, using k-means and hierarchical clustering with different linkage methods on both the

original features and the top principal components. This allows us to then compare the clusters to the actual diagnoses, indicating whether malignant and benign tumors naturally form separate groups. Lastly, we will fit a plain logistic regression as a baseline, then implement ridge and lasso logistic regression, using cross-validation to choose the penalty strength. Lasso will help identify the most important features, while ridge will address the correlated features. We will then compare all three models, focusing on how well they identify malignant tumors.

Evaluation and Interpretation

Comparing results across our methods will play an important role in our evaluation and interpretation. Within each method, we will identify the features that contribute most to separating malignant from benign tumors by looking at group differences in EDA, component loadings in PCA, centroid differences in clustering, and coefficient magnitudes in ridge and lasso. We will then compare these contributions across methods, for example by examining the overlap among the top features, to see whether the same measurements stand out consistently. If the methods agree on the most important features, we can be more confident our results aren't specific to any one technique. Combined with a high rate of correctly identifying malignant tumors and good clustering results, it would give us confidence that malignant and benign tumors can be reliably told apart using these measurements.

Siham: Data prep, EDA, and the cross-method comparison

- Load the data, recode malignant = 1 and benign = 0, standardize the features, and make the stratified train/test split with a fixed random seed. Share this as one script or notebook everyone imports.
- Summary statistics, histograms, boxplots by diagnosis, and the correlation heatmap.
- Flag the highly correlated feature groups, like radius, perimeter and area.
- At the end, gather each person's top features into one comparison table and analyze where they overlap. This is the "Evaluation and Interpretation" section.

Nathan: PCA and k-means

- Fit PCA, make a scree or cumulative variance plot, and interpret the loadings (which features drive PC1 and PC2).
- Plot PC1 vs. PC2 colored by diagnosis.
- Run k-means (k = 2) on the full feature set and on the top PCs, and report which features differ most between the cluster centroids.

Ben: Hierarchical clustering and evaluating all the clusters

- Run hierarchical clustering with single, complete, average and Ward linkage on the full features and on the top PCs, and make dendrograms.
- Compare every clustering result, including Person 2's k-means, to the true diagnoses. Use confusion tables, the adjusted Rand index and purity.
- Write up research question 1: do the groups separate naturally?

Amanda: Classification

- Fit plain logistic regression as the baseline, then ridge and lasso logistic regression with cross-validation to pick the penalty.
- Report recall/sensitivity for malignant tumors, plus precision, accuracy, AUC and confusion matrices on the test set.
- Report the lasso's selected features and the ridge coefficient sizes.
- Write up research question 2.

Shared work

- Each person writes the methods and results sections for their own part.
- Split the intro, dataset description and conclusion, or rotate them.
- Agree early on one random seed, the train/test split, and a format for "top features" (for example, the top 5 by importance) so Person 1 can compare them across methods easily.